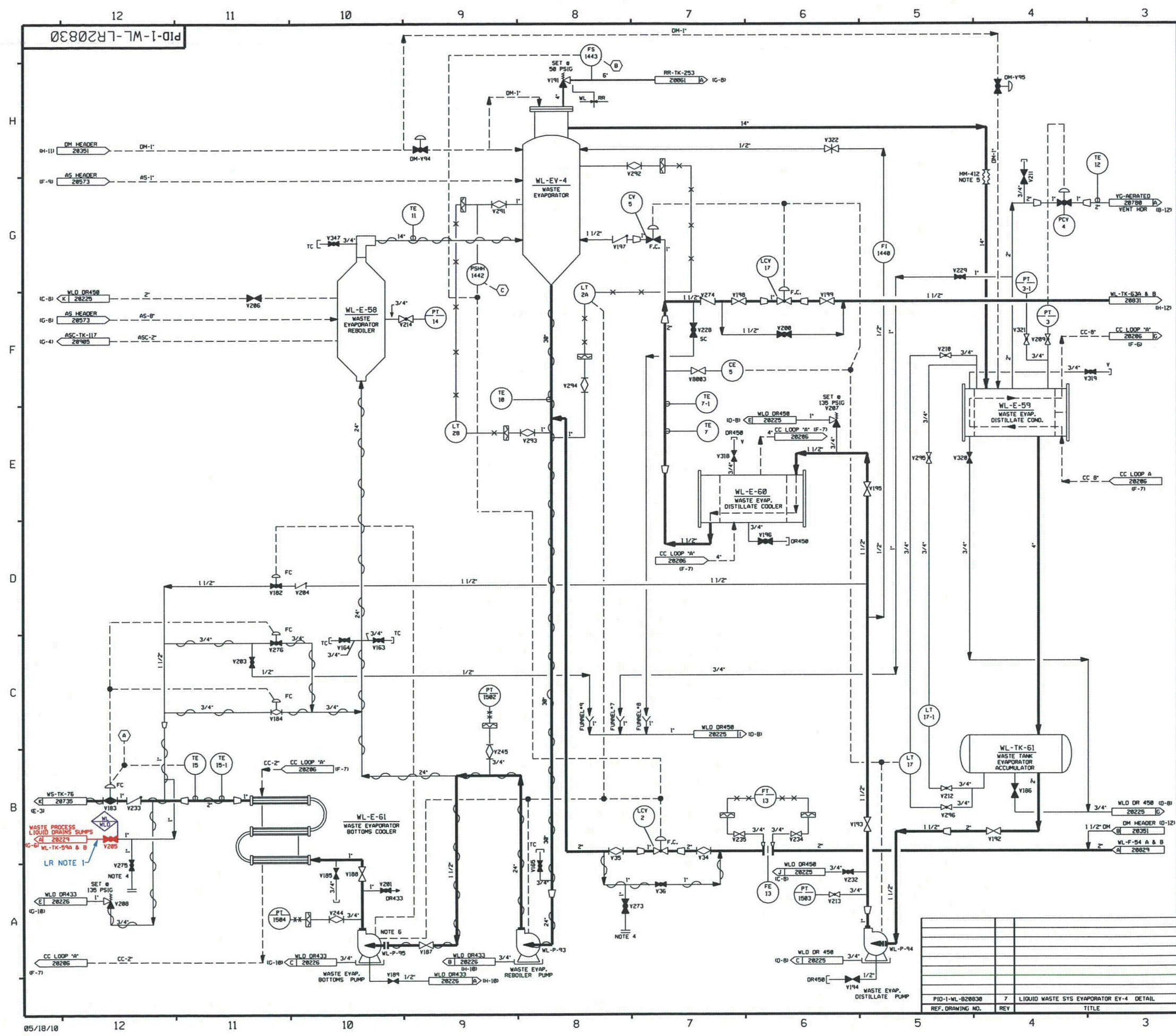




FOR P&ID REFERENCE DRAWINGS, SYMBOLS AND ABBREVIATIONS REFER TO DRAWINGS P&ID - LRLEGEND 1 AND P&ID - LRLEGEND 2

LR NOTES:
1. THE COMPONENTS HIGHLIGHTED IN RED ARE IN LICENSE RENEWAL (LR) SCOPE AS CRITERION (a)(3), FIRE PROTECTION. ALL OTHER COMPONENTS INDICATED ON THIS DRAWING ARE LOCATED IN THE WASTE PROCESSING BUILDING WHICH HAS NO SAFETY RELATED COMPONENTS SO THEY HAVE NO LR INTENDED FUNCTION AND ARE NOT IN SCOPE.

COMPONENTS SUBJECT TO AMR
RED INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(1) AND/OR (a)(3) AND SUBJECT TO AMR PER 10CFR 54.21.
BLACK INDICATES COMPONENTS NOT SUBJECT TO AMR.
GREEN INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(2) AND SUBJECT TO AMR PER 10CFR 54.21.

▲ PIPE ANCHOR
INDICATES SYSTEM BOUNDARY INTERFACE
WLD - WASTE PROCESSING LIQUID DRAINS SYSTEM
WL - LIQUID WASTE SYSTEM

NOTES:
1. WORK THIS DWG WITH DWGS 20828, 20829, & 20831.
2. ALL LINES, EQUIPMENT, COMPONENTS AND INSTRUMENTS ARE SYSTEM PREFIX WL UNLESS NOTED OTHERWISE.
3. ALL LINES, EQUIPMENT, COMPONENTS AND INSTRUMENTS ARE SAFETY CLASS IWS, NON-SEISMIC CATEGORY I UNLESS NOTED OTHERWISE.
4. CHEMICAL ADDITION & REMOVAL CONNECTIONS.
5. ENTIRE LINE IS FLEX CONNECTION.
6. SEE DM-20353 FOR WATER SUPPLY TO PUMP DOUBLE MECHANICAL SEALS.
7. △ - INDICATES REVISION LEVEL.
8. INSTRUMENT REFERENCES:
(A) HIGH WASTE CONCENTRATES TANK VS-TK-75 & HIGH COOLER OUTLET TEMPERATURE & LOW EV-4 LEVEL DISCHARGE INTERLOCK.
(B) C TO 1-DM-Y94 ON DWG DM-20353 04-10, 1-WL-FCY-1 AND 1-AS-V539 ON DWG AS-20573 04-10 AND 07-10.

REV	DATE	DSGN	CHKD	CE	ORIGINAL ISSUE	DESCRIPTION
1						
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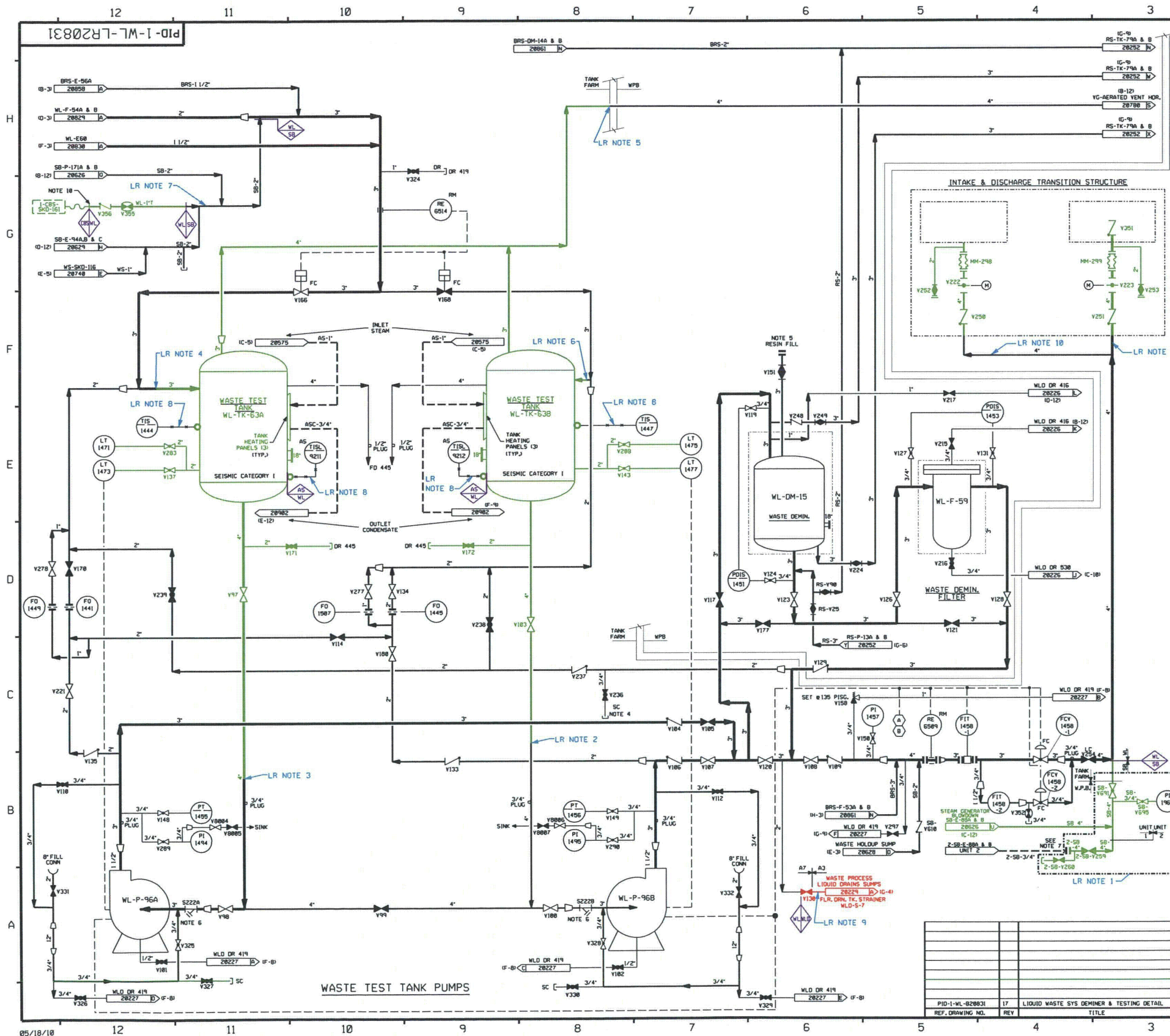
LIQUID WASTE SYSTEM
EVAPORATOR EV-4
DETAIL
LICENSE RENEWAL P&ID

PID-1-WL-LR20830

PID-1-WL-B20830
REF. DRAWING NO.

7
REV

LIQUID WASTE SYS EVAPORATOR EV-4
TITLE



FOR P&ID REFERENCE DRAWINGS, SYMBOLS AND ABBREVIATIONS REFER TO DRAWINGS P&ID - LRLEGEND 1 AND P&ID - LRLEGEND 2

COMPONENTS SUBJECT TO AMR
RED INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(1) AND/OR (a)(3) AND SUBJECT TO AMR PER 10CFR 54.21.
BLACK INDICATES COMPONENTS NOT SUBJECT TO AMR.
GREEN INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(2) AND SUBJECT TO AMR PER 10CFR 54.21.

PIPE ANCHOR
INDICATES SYSTEM BOUNDARY INTERFACE

SB - STEAM GENERATOR BLOWDOWN SYSTEM
WL - LIQUID WASTE SYSTEM
AS - AUXILIARY STEAM SYSTEM
WLD - WASTE PROCESSING LIQUID DRAINS SYSTEM
CBS - CONTAINMENT SPRAY SYSTEM

- NOTES:**
- WORK THIS DRAWING WITH DRAWINGS 20828, 20829 & 20830.
 - ALL LINES, EQUIPMENT, COMPONENTS AND INSTRUMENTS HAVE SYSTEM PREFIX WL UNLESS NOTED OTHERWISE.
 - ALL LINES, EQUIPMENT, COMPONENTS AND INSTRUMENTS ARE SAFETY CLASS NNS UNLESS NOTED OTHERWISE.
 - FLUSH AND SAMPLE CONNECTION USE FLEXIBLE HOSE TO A LOCAL DRAIN FUNNEL TO FLUSH THE LINE, USE POLYETHYLENE COLLECTION BOTTLES TO COLLECT SAMPLES.
 - RESIN FILL TANK (TYP) AS SHOWN ON DWG. BRS-20861.
 - START-UP ENGINEERING TO PROVIDE PIPING, VALVE AND P & ID'S ALSO STRAINER BLOWOFF PIPING & VALVES FOR INITIAL FLUSHING OPERATION, STRAINER SCREEN TO BE REMOVED AFTER START-UP.
 - LINE BLIND FLANGED TO ISOLATE UNIT 1 FROM UNIT 2
 - INDICATES REVISION LEVEL
 - INSTRUMENT REFERENCES
(A) FROM SB-P-242 DRAWING SB-20828
(B) FROM BRS-P-90A&B DRAWING BRS-20861
 - CAP IS INSTALLED WHEN HOSE IS NOT CONNECTED.

- LR NOTES:**
- COMPONENTS SB-V691, SB-V695, PI-1969, 2-SB-V259 AND 2-SB-V260 ARE CONSIDERED IN THE TANK FARM FOR LICENSE RENEWAL (LR).
 - LINE EXISTS TANK FARM AREA DOWNSTREAM OF V-103 (1-WT-942-01).
 - LINE EXISTS TANK FARM AREA DOWNSTREAM OF V-97 (1-WL-941-01).
 - LINE EXISTS TANK FARM AREA PRIOR TO BRANCH CONNECTION TO VALVE V-166 (1-WL-906-11).
 - LINE EXISTS TANK FARM AREA ON ISOMETRIC 1-WL-906-03.
 - LINE EXISTS TANK FARM AREA PRIOR TO BRANCH CONNECTION TO 3" LINE (1-WL-906-02).
 - SKID 1-CBS-SKD-161 IS LOCATED IN THE PRIMARY AUXILIARY BUILDING THE CONNECTING PIPE IS ROUTED TO THE WASTE PROCESSING BUILDING WHICH HAS NO SAFETY RELATED COMPONENTS AND HAS NO LICENSE RENEWAL INTENDED FUNCTION AND NOT IN SCOPE.
 - CAPILLARY TUBING IS PART OF THE INSTRUMENT AND CONSIDERED AN ACTIVE COMPONENT PER NEI 95-10.
 - THE COMPONENTS HIGHLIGHTED IN RED ARE IN LR SCOPE AS CRITERION (a)(3), FIRE PROTECTION.
 - COMPONENTS LOCATED OUTSIDE THE INTAKE & DISCHARGE TRANSITION STRUCTURE ARE NOT IN LR SCOPE SINCE NO SAFETY RELATED COMPONENTS ARE IN THE AREA.

8					ORIGINAL ISSUE	
REV	DATE	DSGN	CHKD	CE	DESCRIPTION	
NEXTera ENERGY						
SEABROOK						
LIQUID WASTE SYSTEM DEMINERALIZATION & TESTING DETAIL LICENSE RENEWAL P&ID						
PID- 1-WL-LR20831						



SEABROOK STATION

PID-1-WT-LR20041

FOR P&ID REFERENCE DRAWINGS, SYMBOLS AND ABBREVIATIONS
REFER TO DRAWINGS P&ID - LRLEGEND 1 AND P&ID - LRLEGEND 2

- LR NOTES:
1. PER OPERATIONS PROCEDURE ON 1029.01, PAGE 42, VALVE 1-WT-V-12 IS CLOSED.
 2. COMPONENTS THAT ARE HIGHLIGHTED IN RED ARE IN LICENSE RENEWAL (LR) SCOPE AS CRITERION (a)(3) FIRE PROTECTION(FP).
 3. COMPONENTS IN BLACK ARE NOT IN SCOPE SINCE THEY HAVE NO LR INTENDED FUNCTION FOR FIRE PROTECTION AND NO (a)(2) REQUIREMENT IN THE FIRE PUMP HOUSE.

COMPONENTS SUBJECT TO AMR

RED INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(1) AND/OR (a)(3) AND SUBJECT TO AMR PER 10CFR 54.21.

BLACK INDICATES COMPONENTS NOT SUBJECT TO AMR.

GREEN INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(2) AND SUBJECT TO AMR PER 10CFR 54.21.

- ▲ PIPE ANCHOR
- ◇◇◇◇ INDICATES SYSTEM BOUNDARY INTERFACE
- FP - FIRE PROTECTION SYSTEM
- WT - WATER TREATMENT SYSTEM

- NOTES:
1. WORK THIS DRAWING WITH DRAWINGS 20036, 20038, 20039, AND 20040
 2. ALL LINES, EQUIPMENT, COMPONENTS AND INSTRUMENTS HAVE SYSTEM PREFIX WT, UNLESS NOTED OTHERWISE.
 3. ALL LINES, EQUIPMENT, COMPONENTS AND INSTRUMENTS ARE SAFETY CLASS NMS, UNLESS NOTED OTHERWISE.
 4. △ - INDICATES REVISION LEVEL
 5. DELETED

REV	DATE	DSGN	CHKD	CE	ORIGINAL ISSUE
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05/18/10



SEABROOK STATION

WATER TREATMENT
DETAIL
LICENSE RENEWAL P&ID

PID-1-WT-LR20041

